

MATHEMATICAL FOUNDATIONS & DATA SCIENCE

Introduction to Probability & Statistics

A Comprehensive Overview of Randomness, Data Modeling, Descriptive Measures, and
Inferential Reasoning

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Part I: Fundamentals of Probability

Understanding random experiments, sample spaces, axiomatic probability, and conditional reasoning.

Foundations of Probability

Core Concepts & Axioms

Probability quantifies the likelihood of occurrence for uncertain events in a random experiment.

- **Sample Space (S):** The set of all possible outcomes of an experiment.
- **Event (A):** Any subset of the sample space S.
- **Non-negativity:** $P(A) \geq 0$ for any event A.
- **Normalization:** $P(S) = 1$ for the total sample space.

$$P(A) = \frac{n(A)}{n(S)}$$

Approaches to Probability

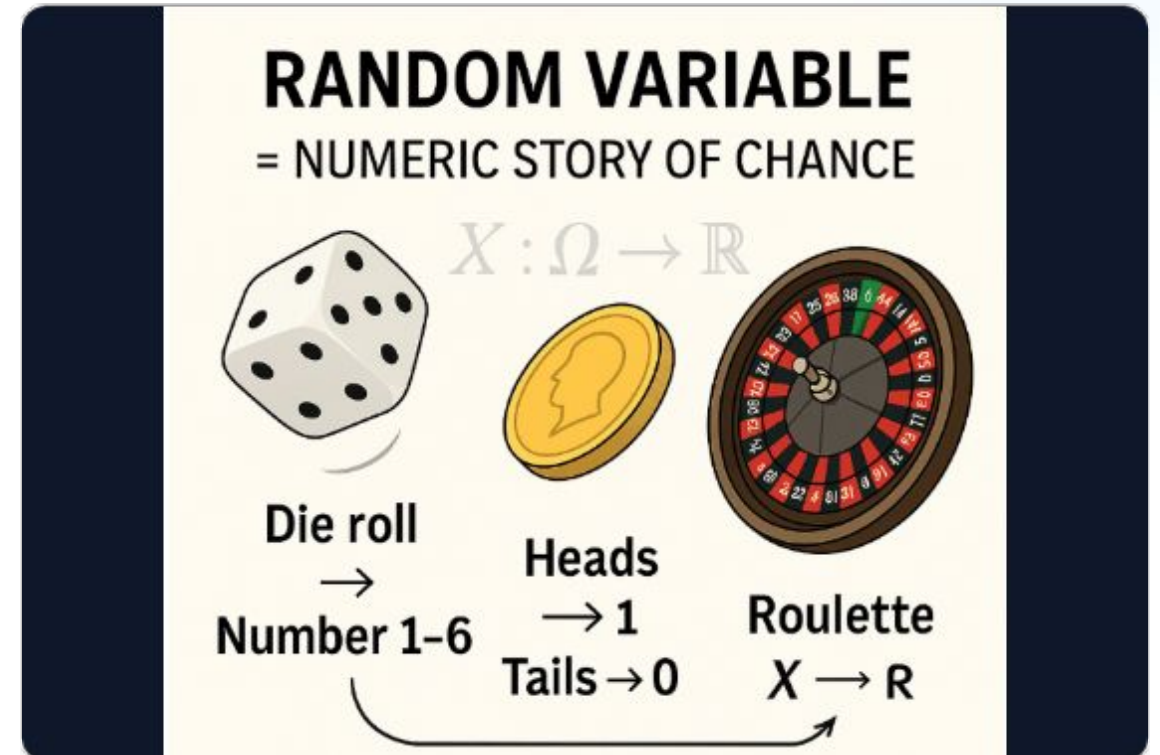
Three main frameworks are used to interpret and calculate probabilities in practice:

- **Classical Probability:** Based on equally likely outcomes (e.g., fair dice or coin flips).
- **Empirical (Relative Frequency):** Based on historical observation and experiment trials.
- **Subjective Probability:** Based on expert opinion, belief, or prior intuition.

Probability Rules & Bayes' Theorem

- **Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. For mutually exclusive events, $P(A \cap B) = 0$.
- **Conditional Probability:** Likelihood of event A given that event B has occurred.
- **Multiplication Rule:** $P(A \cap B) = P(A|B) \cdot P(B)$. For independent events, $P(A \cap B) = P(A) \cdot P(B)$.
- **Bayes' Theorem:** Revises prior probabilities based on new observed evidence.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$



Common Probability Distributions

Distribution	Type	Key Parameters	Typical Application
Binomial	Discrete	n (trials), p (probability)	Number of successes in n independent Bernoulli trials
Poisson	Discrete	λ (rate parameter)	Count of rare events occurring in a fixed time interval
Uniform	Continuous / Discrete	a (min), b (max)	Events where all outcomes are equally likely to occur
Normal (Gaussian)	Continuous	μ (mean), σ (std dev)	Natural phenomena, measurement errors, standard scores

Part II: Descriptive Statistics

Summarizing, organizing, and describing data using numerical measures and visual distributions.

Central Tendency & Dispersion



Central Tendency

Identifies the center or typical value of a dataset:

- **Mean (μ):** Arithmetic average of all observations.
- **Median:** Middle value when data is sorted sequentially.
- **Mode:** Most frequently occurring value in data.



Measures of Spread

Quantifies the variability or dispersion in data:

- **Range:** Difference between maximum and minimum values.
- **Variance (σ^2):** Average squared distance from the mean.
- **Std Deviation (σ):** Square root of variance in original units.



Shape of Data

Describes distribution asymmetry and peakedness:

- **Skewness:** Measure of symmetry (positive or negative tail).
- **Kurtosis:** Measure of heavy tails or sharpness of peak.

Empirical Rule of Normal Distribution

99.7%

Data within 3 Standard Deviations

The 68–95–99.7 Rule

For any bell-shaped normal distribution curve:

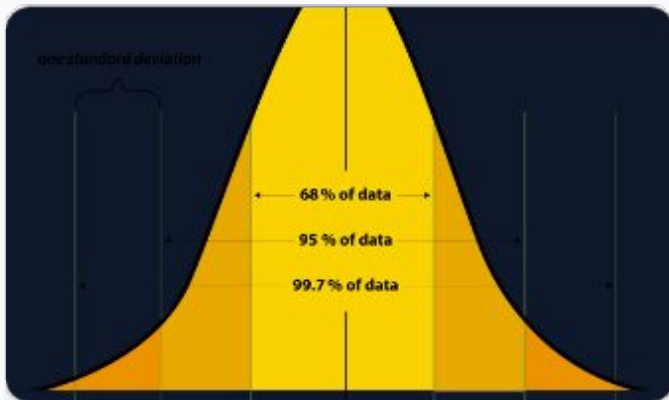
- 68.2% of data falls within 1 standard deviation ($\mu \pm 1\sigma$).
- 95.4% of data falls within 2 standard deviations ($\mu \pm 2\sigma$).
- 99.7% of data falls within 3 standard deviations ($\mu \pm 3\sigma$).

$$Z = \frac{x - \mu}{\sigma}$$

Part III: Inferential Statistics

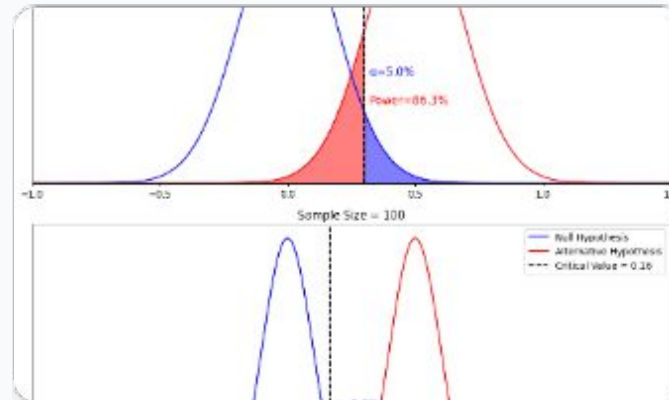
Drawing conclusions, making predictions, and testing hypotheses about populations using sample data.

Core Branches of Inferential Analysis



Estimation & Confidence Intervals

Uses sample statistics to estimate population parameters within a calculated margin of error.



Hypothesis Testing

Tests claims about populations using Null (H_0) and Alternative (H_1) hypotheses with p-values.



Regression & Modeling

Evaluates mathematical relationships between dependent and independent variables for forecasting.

Statistical Analysis Workflow

